A neural network potential for the Zundel ion

Lesson notes accompanying `tutorial_01_zundel/zundel.ipynb`

Machine-learned interatomic potentials — hands-on session

Abstract

These notes go with the first notebook of the MLIP tutorial. You will build a machine-learned interatomic potential (MLIP) from start to finish: reference data, descriptors, a neural network, active learning by *query by committee*, production molecular dynamics (MD) in CP2K, and validation against the reference. The system is deliberately tiny: one $H_5O_2^+$ ion in vacuum, two chemical elements, a semi-empirical reference method and a network with ten hidden neurons. It is a **basic, pedagogical example**. Every step is the same one you would take in a research project, but every number is chosen so that the whole workflow runs on a laptop in a few minutes. The last section lists the subtleties that can be added on top of this simple model, and how you would tackle the same problem in a research setting.

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1 Why machine-learned potentials?

An MD simulation integrates Newton's equations $m_i\ddot{\mathbf{r}}_i = \mathbf{F}_i = -\nabla_{\mathbf{r}_i} E(\{\mathbf{r}\})$. Everything depends on the potential energy surface (PES) $E(\{\mathbf{r}\})$, and there are two traditional ways to get it:

- **Electronic structure** (DFT, quantum chemistry, or here DFTB). It is accurate and reactive, but costs roughly $\mathcal{O}(N^3)$ or more per step. This limits you to $\sim 10^2$–10^3 atoms and tens of picoseconds.
- **Empirical force fields**. They cost $\mathcal{O}(N)$ and are very fast, but have a fixed functional form. They usually cannot break bonds and are only as accurate as their parametrisation.

An MLIP is a flexible regression model $E_\theta(\{\mathbf{r}\})$ fitted to reference electronic-structure energies and forces. It has the $\mathcal{O}(N)$ cost of a force field, and *within the region of configuration space covered by its training data* it reaches the accuracy of the reference. The italicised part is the whole difficulty, and it is the thread running through this tutorial.

What this example is — and what it is not

System	One $H_5O_2^+$ ion (7 atoms) in vacuum, no periodic boundaries.
Reference	SCC-DFTB, NVT at 300 K, 25 ps trajectory. Cheap, but *not* a high-accuracy method.
Model	2nd-generation Behler–Parrinello network [1]: one 28/32 → 5 → 5 → 1 network per element, about 380 weights in total.
Training	n2p2 [3] (Kalman filter), driven by the `aml` package [5].
MD	CP2K [8], NNP force environment.

Research-grade MLIPs are trained on DFT or coupled-cluster data, for condensed phases with thousands of atoms, often with many elements. They use much larger (often equivariant, message-passing) models, or start from a pretrained foundation model (Sec. 8). The *concepts* you practise here — locality, descriptors, committees, active learning, validation by observables, transferability — carry over unchanged.

2 The system and the reference data

The Zundel ion $H_5O_2^+$ is an excess proton shared between two water molecules, $[H_2O{\cdots}H{\cdots}OH_2]^+$. Together with the Eigen cation $H_9O_4^+$ it is one of the two limiting structures of the hydrated proton. It is central to the Grotthuss mechanism of proton transport in water [9]. Its PES is strongly anharmonic: the shared proton moves in a very flat, nearly symmetric potential along the O–O axis. Even this small cluster is therefore a non-trivial test for a potential.

Reference trajectory (`data/`).

- Method: self-consistent-charge DFTB [10] in CP2K, total charge +1.
- Ensemble: NVT, 300 K, CSVR thermostat [11] ($\tau = 200$ fs), $\Delta t = 0.5$ fs, 50 000 steps (25 ps).
- Output every 5 steps: 10 001 frames, 2.5 fs apart, with positions, *forces* and velocities, plus the energy file `zundel-1.ener`.
- Centre-of-mass translation and rotation are removed.

Every frame is a labelled data point $(\{\mathbf{r}\}, E, \{\mathbf{F}\})$. One structure with 7 atoms gives 1 energy but 21 force components, so the forces carry most of the information.

Q. *Consecutive frames are only 2.5 fs apart. How many of the 10 001 frames are really* independent*? What does that imply for a random train/test split?*

3 The Behler–Parrinello neural network potential

3.1 Locality: energy as a sum of atomic energies

The central approximation of almost all MLIPs is *locality*. The total energy is a sum of atomic contributions, and each one depends only on the neighbours within a cutoff radius r_c:

$$E(\{\mathbf{r}\}) = \sum_{i=1}^{N} E_{Z_i}(\mathbf{G}_i)\,, \qquad \mathbf{F}_j = -\sum_i \frac{\partial E_{Z_i}}{\partial \mathbf{G}_i} \cdot \frac{\partial \mathbf{G}_i}{\partial \mathbf{r}_j}. \tag{1}$$

E_Z is a feed-forward network, one per chemical element Z, shared by all atoms of that element. The construction makes the cost linear in N and the model size-extensive: you can, in principle, apply it to a larger system (Sec. 7). The forces are *exact* derivatives of the energy, so the model is conservative by design.

3.2 Descriptors: atom-centred symmetry functions

The input $\mathbf{G}_i$ must be invariant under translation, rotation and permutation of identical atoms. Behler–Parrinello models use atom-centred symmetry functions (ACSFs) [2]. With the cosine cutoff $f_c(r) = \frac{1}{2}\left[\cos(\pi r/r_c) + 1\right]$ for $r < r_c$:

$$G_i^{\mathrm{rad}} = \sum_{j\neq i} e^{-\eta(r_{ij}-r_s)^2} f_c(r_{ij}), \tag{2}$$

$$G_i^{\mathrm{ang}} = 2^{1-\zeta} \sum_{j,k\neq i} (1 + \lambda\cos\theta_{ijk})^{\zeta}\, e^{-\eta(r_{ij}^2+r_{ik}^2+r_{jk}^2)} f_c(r_{ij}) f_c(r_{ik}) f_c(r_{jk}). \tag{3}$$

Each function is resolved by the elements of the neighbours, so there are separate functions for O–O and O–H pairs, and for each neighbour pair in the angular terms. The `aml` default set used in the notebook has:

- cutoff $r_c = 12\,a_0 \approx 6.35$ Å, which covers the whole ion;
- 10 radial functions per element pair, with Gaussians shifted between 0.5 Å and $r_c - 0.5$ Å;
- 4 angular functions per triple ($\lambda = \pm 1$, $\zeta = 1, 4$, centred);
- the O–O–O triple excluded (`exclude_triples`), since the ion has only two oxygens.

This gives 28 descriptors for O and 32 for H. You can plot them with `aml.acsf.plot_radial` and `aml.acsf.plot_angular`.

3.3 Network and training

The notebook defines each element network in `input.nn` through `set_nn()`: 2 hidden layers of 5 neurons, tanh activations and a linear output (`t t l`). That is 181 weights for O and 201 for H. n2p2 fits energies *and* forces with a (global) extended Kalman filter [4], a second-order optimiser that is very efficient for small networks. 10 % of the structures are held out as a test set, and the `learning-curve.out` files record the train/test RMSE per epoch.

Units

`aml` works internally in atomic units (Hartree, Bohr). The notebook converts with $1\,E_h/a_0 = 51.422\,\mathrm{eV/Å}$ and $1\,E_h = 27.211\,\mathrm{eV}$. Force errors are usually quoted in meV/Å.

4 A naive first model

The first experiment trains a *single* network on every 25th frame of the reference trajectory: 401 structures, evenly spaced in time, for 50 epochs. The test errors look reasonable. You then run MD with this model (`single/`, 50 ps).

In a test run of this notebook, the dynamics stayed stable for about 4.9 ps. The temperature then jumped above 1000 K, reached $\sim 10^5$ K, and the ion dissociated: the O–O distance, which stays between 2.2 and 2.7 Å in the reference, grew to $\sim$100 Å. The details change from run to run (random initial weights, random train/test split), but a failure of this kind is typical.

Why? The MD trajectory has wandered into a region of configuration space the training set did not cover: a rare, high-energy fluctuation of the shared proton or of the O–O distance. There the network *extrapolates*. A neural network has no physical prior, so it can predict an unphysical hole in the PES that pulls atoms together, or a spurious force that pumps in energy. Once the system enters such a region, the error feeds on itself.

Lesson 1: a low test RMSE does not guarantee stable MD

The test set comes from the same distribution as the training set, so it cannot detect holes outside that distribution. MD finds them, because it explores configuration space systematically, including its rare-event tails. This observation has since been documented systematically [12]. Always validate a potential by *running* it.

Q. *Evenly spaced frames sample configurations by their Boltzmann weight. Which configurations are therefore under-represented, and why are they exactly the dangerous ones?*

5 Committees and query by committee

5.1 The committee as an uncertainty estimate

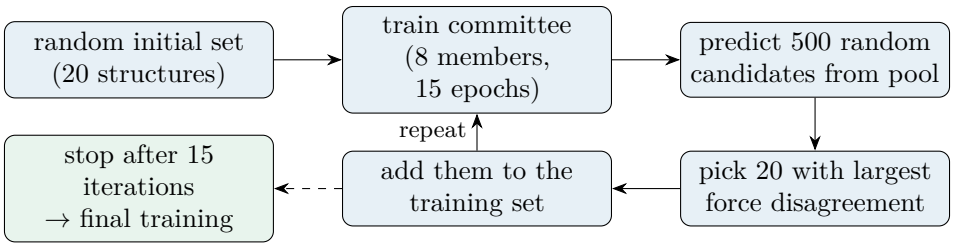
Instead of one network, train a *committee* of n networks (here $n = 8$). They differ in initial weights, in the random train/test split and in the order of the stochastic updates [5]. The prediction is the committee mean, and the spread is an uncertainty estimate:

$$\bar{E} = \frac{1}{n}\sum_{k=1}^{n} E^{(k)}, \qquad \sigma_E = \sqrt{\frac{1}{n}\sum_{k=1}^{n}\left(E^{(k)} - \bar{E}\right)^2}, \tag{4}$$

and likewise for every force component. Where training data exist, the members agree. Away from the data each one extrapolates differently, and σ grows. A large committee disagreement is not an error bar in the strict statistical sense, but it is a cheap and remarkably useful indicator of *"this configuration is outside what I know"*.

5.2 Active learning: query by committee (QbC)

Query by committee [7] uses the disagreement to choose training data. The notebook applies it to an existing pool (the full 10 001-frame reference trajectory), so no new reference calculations are needed:



For each candidate, `aml` computes the committee standard deviation of every force component. The structure is then scored by a high-order (6th) moment of these values, which emphasises the single worst atom rather than the average. After 15 iterations the training set holds at most

$20 + 15 \times 20 = 320$ structures, *fewer* than the 401 of the naive model. The final model is a fresh 8-member committee, trained for 50 epochs on this set (`train/final_training`).

The notebook prints three disagreements at every iteration: on the training set, on the random candidates, and on the selected structures. Track how they evolve. In a test run (Table 1), the disagreement on the candidates fell by a factor of $\sim$20 over 15 iterations. Late iterations started re-selecting structures already in the set, a sign of convergence. The final model was trained on 284 structures.

Table 1: Mean committee force disagreement (meV/Å) during QbC in a test run. Your numbers will differ (random sampling and weights); the trend will not.

iteration	training-set size	train	candidates	selected
1	20	307	695	5609
2	40	152	151	289
3	60	111	95	139
5	100	85	70	108
10	194	62	47	68
15	284	51	37	56

Q. *Is the disagreement on the training set always the smallest? Check Table 1: from iteration 2 on it is* larger *than on random candidates. Why? (Think about which structures the training set is made of.) Why do the selected structures sit preferentially at* high *potential energy along the trajectory (plot of training points vs. $E_p(t)$)?*

Q. *How does the wall time of one QbC iteration grow with the iteration index? (In the test run it barely changed, 11–12 s: for such small sets the fixed costs dominate.) Which part of the work scales with the training-set size, and which with the committee size?*

Lesson 2: not all configurations are equally valuable

A thermal trajectory is extremely redundant. Active learning spends the reference-data budget on the configurations the model is most uncertain about. This gives a smaller and better training set, and in production it lets the uncertainty be monitored on the fly.

6 Production MD and validation

6.1 Running the committee in CP2K

`N2P2.save_model()` writes one directory `nnp-XXX` per member. The CP2K input loads all of them:

```
@SET PATH ../train/final_training/model
&NNP
  NNP_INPUT_FILE_NAME ${PATH}/nnp-000/input.nn
  SCALE_FILE_NAME     ${PATH}/nnp-000/scaling.data
  &MODEL
    WEIGHTS ${PATH}/nnp-000/weights
  &END MODEL
  ...                        # one &MODEL block per member
&END NNP
```

CP2K propagates the dynamics with the committee mean. The `&PRINT/&ENERGIES` section writes the energy of every member, from which you get the committee disagreement along the trajectory. The cost grows linearly with the number of members. If you change `n_members`, edit the `&MODEL` blocks to match.

6.2 Validation by observables

Low force errors are necessary but not sufficient, so the notebook compares *simulation observables* between the NNP and DFTB trajectories with `aml.score`:

- **Radial distribution functions** $g_{\alpha\beta}(r)$ for the O–O, O–H and H–H pairs (structure).
- **Vibrational density of states**, computed from the velocity autocorrelation function, $I(\omega) \propto \int \sum_i \langle \mathbf{v}_i(0) \cdot \mathbf{v}_i(t) \rangle \, w(t) \, e^{-i\omega t} \, dt$, with a Hann window w (dynamics). The O–H stretch region near $3000\,\mathrm{cm}^{-1}$ and the low-frequency shared-proton modes are the sensitive parts.

For each curve the score is a normalised absolute deviation, $\varepsilon = \sum |f_{\mathrm{ref}} - f_{\mathrm{NNP}}| / (\sum f_{\mathrm{ref}} + \sum f_{\mathrm{NNP}})$, where 0 means identical, and `aml` prints $100\,(1-\varepsilon)$ as an "accuracy" in %. In the test run, the final committee ran 50 ps without problems: the temperature stayed below 750 K, as in the reference, and the O–O distance stayed in 2.22–2.63 Å (reference 2.24–2.69 Å). The scores were 95.8 % for the RDFs (O–O 92.8 %, O–H 97.0 %, H–H 97.5 %) and 90.6 % for the VDOS.

Q. *Why is the O–O RDF error typically the largest? (Count how many O–O pairs contribute to the histogram.) Is it surprising to reach good agreement for a single molecule?*

7 Transferability: a bigger cluster

Thanks to Eq. (1), nothing stops you from running the same model on $(H_2O)_6H^+$ (`cluster/`, 500 fs). Running it is technically possible; getting it right is not guaranteed. The larger cluster contains environments the Zundel ion never had:

- water–water hydrogen bonds that do not involve the excess proton;
- more neighbours inside the cutoff, so descriptor values fall outside the range seen in training (`scaling.data` records that range);
- O–O–O triples, whose angular terms were *excluded* from the model.

In the test run the temperature exceeded 1000 K within the first femtoseconds and reached $\sim 10^5$ K: the model fails immediately. Use the committee disagreement to diagnose such failures. This motivates the second tutorial (`wclusters.ipynb`), which trains on the larger cluster and probes transfer to a 17-water cluster and to bulk water.

Lesson 3: an MLIP knows only what it has seen

Locality makes a model *applicable* to new systems; only training data make it *reliable* there. Transferability must be tested, never assumed. And the model can never be better than its reference: as tutorial 2 shows, this DFTB parametrisation gives heavily over-structured liquid water.

8 Going further: subtleties beyond the simple model

The workflow above — Behler–Parrinello networks, ACSFs, committee-based active learning — is deliberately simple, and perfectly valid. It is a very good way to *understand* MLIPs. Each of its ingredients can be refined, and the subtleties below are the main ones a practitioner would add on top of it.

8.1 Architectures: from fixed descriptors to equivariant message passing

- **Systematic many-body descriptors.** The atomic cluster expansion (ACE) [13] provides a complete, systematically improvable basis that replaces hand-chosen ACSF sets.
- **Equivariant message-passing networks.** NequIP [14], Allegro [15] and MACE [16] learn the representation instead of fixing it. They propagate geometric (tensorial) features that

rotate with the structure, and capture many-body correlations efficiently. They are markedly more data-efficient than descriptor-based networks: a few hundred structures can be enough where thousands were needed before [14].

- **Long-range interactions.** The strict cutoff in Eq. (1) misses electrostatics and charge transfer, which matters for an *ion* like $H_5O_2^+$. Fourth-generation networks [17] learn environment-dependent charges with a global charge equilibration. Latent Ewald summation [18] adds a learned long-range term without reference charges.
- **Conservative vs. direct forces.** Some large models predict forces directly instead of as $-\nabla E$, for speed. This was shown to cause drift and instabilities in MD [19], so conservative forces are preferred, at least for simulation [20]. The Behler–Parrinello model you train here is conservative by construction.

8.2 Foundation (universal) models

Instead of fitting *one potential per system* from scratch, as done here, one can start from a *pretrained general-purpose potential*:

- **Materials.** MACE-MP-0 [21], trained on the Materials Project relaxation trajectories, runs stable MD for most of the periodic table out of the box. Meta's OMat24 [22] dataset ($> 10^8$ DFT calculations) and the models trained on it pushed accuracy further.
- **Molecules.** MACE-OFF [23] targets organic molecules. OMol25 [24] contains $> 10^8$ ωB97M-V/def2-TZVPD calculations, including *charged and open-shell* systems, explicit solvation and reactive structures.
- **One model for everything.** UMA [25] trains a single model family jointly on materials, molecules, catalysis and metal–organic-framework data.
- **Fine-tuning.** A foundation model can be fine-tuned on a small, system-specific set (tens to hundreds of structures) to reach the accuracy of a chosen reference [21]. The active-learning ideas of this tutorial then serve to *pick those fine-tuning structures*.

8.3 Evaluation and uncertainty

- **Beyond RMSE.** A low RMSE is not enough: benchmarks [12] also judge models by MD stability, energy conservation [20] and physical observables. This is exactly the philosophy of the RDF/VDOS scoring in this notebook.
- **Committees are the workhorse.** Committees/ensembles, as used here [5], are the most common and robust way to estimate uncertainty and drive active learning. The automated QbC workflow was later applied to complex aqueous interfaces [6] (tutorial 3).

How would you do this exercise in a research project?

1. Generate reference data with a better method than DFTB, e.g. a hybrid functional with dispersion. For this ion, which carries a net charge, the method should handle a charged system properly.
2. Start from a pretrained model whose training data cover *charged* molecules (e.g. one trained on OMol25). Check it on a few reference frames before trusting it.
3. Fine-tune an equivariant model (e.g. MACE) on a few dozen to a few hundred frames, selected by committee or ensemble disagreement, exactly as in Sec. 5.
4. Validate with observables (RDF, VDOS, proton-sharing coordinate $\delta = r_{O_1H^*} - r_{O_2H^*}$) and test MD stability over long runs.
5. Consider nuclear quantum effects (path-integral MD). For the shared proton in $H_5O_2^+$ they are *large*, and the cheap MLIP is what makes them affordable.

9 Running times

Measured on an 8-core workstation (one CPU core per committee member):

step	wall time
naive model: training (1 member, 401 structures, 50 epochs)	11 s
naive model: CP2K MD (100 000 steps, 50 ps)	2.4 min
QbC (15 iterations, 8 members)	3.0 min
final training (8 members, 284 structures, 50 epochs)	18 s
committee MD in CP2K (100 000 steps, 50 ps)	2.6 min
7-water cluster MD (1000 steps)	5 s

On a laptop with fewer cores, QbC and committee training take proportionally longer (the notebook runs 8 members in parallel in QbC and 4 in the final training). The first CP2K run can be started while you read the next section.

10 Exercises

The blue questions in the notebook, collected:

1. After the naive model's MD: *what is happening?* Propose a few hypotheses and test one (e.g. compare the O–O distance and the potential energy with the reference ranges).
2. Explain the differences in committee disagreement between the selected, training and candidate sets.
3. How does the compute time grow with QbC iterations? What can you say about the computational complexity?
4. Why do the selected training points lie at high potential energy?
5. Change the architecture (`n_hidden`, `n_neurons`, `activ`) or the number of committee members. How do accuracy and MD cost change? (Remember to edit the CP2K `&MODEL` blocks.)
6. Compare the training-set sizes of the 1-member and 8-member models. Are all configurations equally valuable?
7. Run the 7-water cluster. What happens, and why?

Going further. Tutorial 2 (`wclusters.ipynb`) applies the same pipeline to $(H_2O)_6H^+$, tests transfer to $(H_2O)_{16}H^+$ and bulk water, and questions the DFTB reference itself. Tutorial 3 (`wsystems.ipynb`) uses published research-grade committee models for complex aqueous systems [6].

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